

Older Lenia organisms resist the same disturbance

16–29 August 2026 · Developmental perturbation study.

We often describe an organism as becoming more stable as it develops, but size alone can create the same appearance. A fixed disturbance spread across a larger body may simply be more dilute. To distinguish growth from a deeper change in organization, we fixed the experiment before opening the outcomes and applied matched disturbances to younger and older members of 16 Flow Lenia families.

The disturbance touched about 12 per cent of the organism's matter at both ages. Even with that proportion held steady, its effect after four simulation steps was 91 per cent smaller in the older organisms.

We then increased the dose until the younger and older bodies showed a comparable initial disruption. The older bodies required 7.7 times as much. Development had changed the body's susceptibility: the matched intervention no longer had the same power to alter its state.

A companion experiment asked whether a controller could exploit a view of the whole organism to steer it better than a local strategy could. While control was active, the whole-body controller outperformed leaving the organism alone. Once "do nothing" was included among the strategies available for each individual case, however, whole-body control was not reliably better than the safest local alternative. At 128 steps after control ended, its advantage had reversed. The controller could force a stronger short-term correction without leaving behind the most durable recovery.

Separate experiments asked whether a response could identify its source and whether hidden channel composition could transfer that response between bodies. Both an aligned donor transplant and a spatially scrambled version moved the recipient toward the donor's response. The difference between those treatments was unresolved at both tested ages. Changing hidden composition affected the outcome, but these controls did not establish transfer of a response specific to the donor's arrangement.

Taken together, the results show that development changed more than body size. Older organisms responded less to the matched disturbance at the measured early horizon. Whole-body control improved short-term correction relative to leaving the body alone, but its advantage did not survive the strongest local comparison and release test. Recognizing a body by its response and transferring that response to another body remain different experimental questions.

Source: Specter research note N-014, 16–29 August 2026. Journal edition of the web research note.